These spores are just cells of the plant like any others. As in the case of splurge-weed, no sex is involved. The daughters of a plant consist of cells that are clone-mates of the cells of the parent plant. The only difference between the two species is that splurge-weed reproduces by hiving off chunks of itself consisting of indeterminate numbers of cells, while bottle-wrack reproduces by hiving off chunks of itself always consisting of single cells.

By imagining these two kinds of plant, we have zeroed in on the crucial difference between a bottlenecked and an unbottlenecked life cycle. Bottle-wrack reproduces by squeezing itself, every generation, through a single-celled bottleneck. Splurge-weed just grows and breaks in two. It hardly can be said to possess discrete 'generations', or to consist of discrete 'organisms', at all. What about bottle-wrack? I'll spell it out soon, but we can already see an inkling of the answer. Doesn't bottle-wrack already seem to have a more discrete, 'organismy' feel to it?

Splurge-weed, as we have seen, reproduces by the same process as it grows. Indeed it scarcely reproduces at all. Bottle-wrack, on the other hand, makes a clear separation between growth and reproduction. We may have zeroed in on the difference, but so what? What is the significance of it? Why does it matter? I have thought a long time about this and I think I know the answer. (Incidentally, it was harder to work out that there was a question than to think of the answer!) The answer can be divided into three parts, the first two of which have to do with the relationship between evolution and embryonic development.

First, think about the problem of evolving a complex organ from a simpler one. We don't have to stay with plants, and for this stage of the argument it might be better to switch to animals because they have more obviously complicated organs. Again there is no need to think in terms of sex; sexual versus asexual reproduction is a red herring here. We can imagine our animals reproducing by sending off nonsexual spores, single cells that, mutations aside, are genetically identical to one another and to all the other cells in the body.

The complicated organs of an advanced animal like a human or a woodlouse have evolved by gradual degrees from the simpler organs of ancestors. But the ancestral organs did not literally change themselves into the descendant organs, like swords being beaten into ploughshares. Not only *did* they not. The point I want to make is that in most cases they